



SQL-Mongo Project – IBM HR Analytics Employee Attrition & Performance

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Relational Data Model

Initial Assumptions about Entities & their Relationships:

Data Modelling

- The given IBM employee attrition dataset has been partitioned into different entities (logical groupings of data elements) containing attributes and unique identifiers (primary key)
- The relationships between them (1:1, 1: Many, Many: Many) have also been established.

Entities & their attributes:

1. **employee_demographic**: EmployeeNumber, Age, Over18, DistanceFromHome, Gender, MaritalStatus, Attrition, BusinessTravel, WorkLifeBalance, JobInvolvement, OverTime, JobLevel, JobRole, JobRole_des, Department, StandardHours

2. **employee_education**: EmployeeNumber (PK & FK), Education, EducationField

3. **work_history**: EmployeeNumber (PK & FK), NumCompaniesWorked, TotalWorkingYears, TrainingTimesLastYear

4. **current_company_details**: EmployeeNumber (PK & FK), YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager

5. **wages**: EmployeeNumber (PK & FK), HourlyRate, DailyRate, MonthlyRate, MonthlyIncome, PerformanceRating, PercentSalaryHike

6. **company_benefits** : EmployeeNumber (PK & FK), JobSatisfaction, RelationshipSatisfaction, StockOptionLevel, EnvironmentSatisfaction

'employee_demographic' being the fact table every other table has **one to one relationship** with them. All the entities are having identifying relationships and participation is mandatory.

Note: Primary key is denoted as PK & Foreign Key as FK.

Normalization Process:

- This process allows you to filter away unnecessary data from your database.
- Restructuring the tables is necessary to gradually satisfy increasing levels of normalization.

Various levels of normalization are:

- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Boyce-Codd Normal Form (BCNF)
- Fourth Normal Form (4NF)
- Fifth Normal Form (5NF)
- Domain Key Normal Form (DKNF)

1NF: First Normal Form

- All the tables under 'ibmattritiondb' have attributes with atomic value – single scalar value (No multi-valued or multi-part values)
- Every entity has a single identifier (Primary key) that can uniquely identify each instance of the entity

2NF: Second Normal Form

- Attributes that don't depend on the entire primary key is moved to another table and a relationship between the two tables is established.
- All attributes that is not primary key, or non-key attribute are ensured to be functionally dependent on the entire primary key.

PercentSalaryHike Attribute in 'wages' entity depends only on the PerformanceRating Attribute. Hence, it is moved to a separate table called '**performance**' as PerformanceRating (PK & FK), PercentSalaryHike

One to Many Relationship has been established between '**performance**' & '**wages**' table

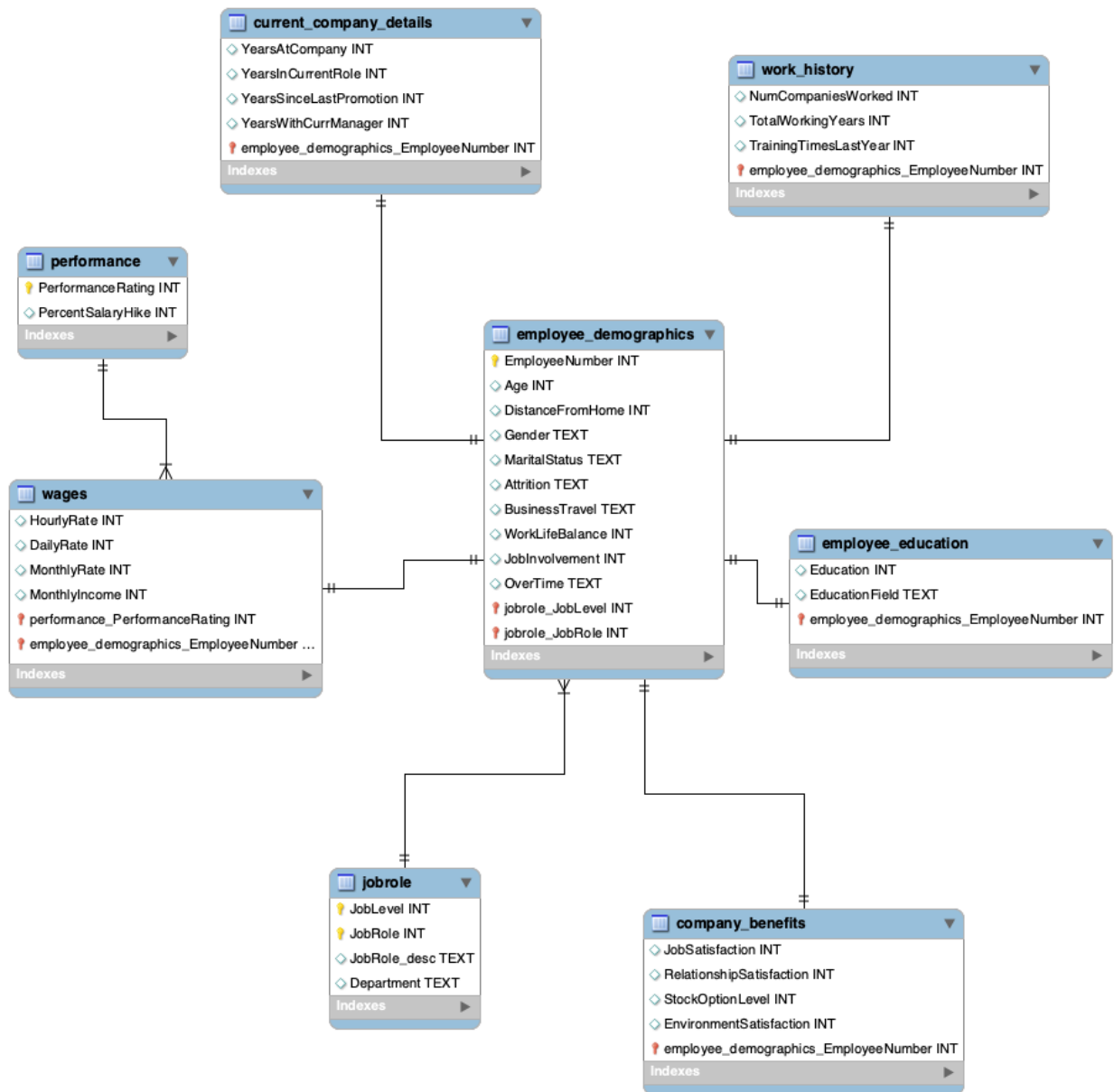
Also, JobRole_des & Department Attributes can be uniquely identified by Joblevel & JobRole attributes. Hence, a separate table '**jobrole**' is created with Joblevel(PK & FK), JobRole (PK & FK), JobRole_des & Department

One to Many Relationship has been established between **jobrole** & **employee_demographics** table

3NF: Third Normal Form

- To ensure there is no Transitive Dependency
- All attributes are ensured to be functionally dependent ONLY on the complete primary key of their entity and not on any other non-key attribute.

Entity-Relationship Diagram



Physical MySQL Database

Assumptions/Notes About Data Set

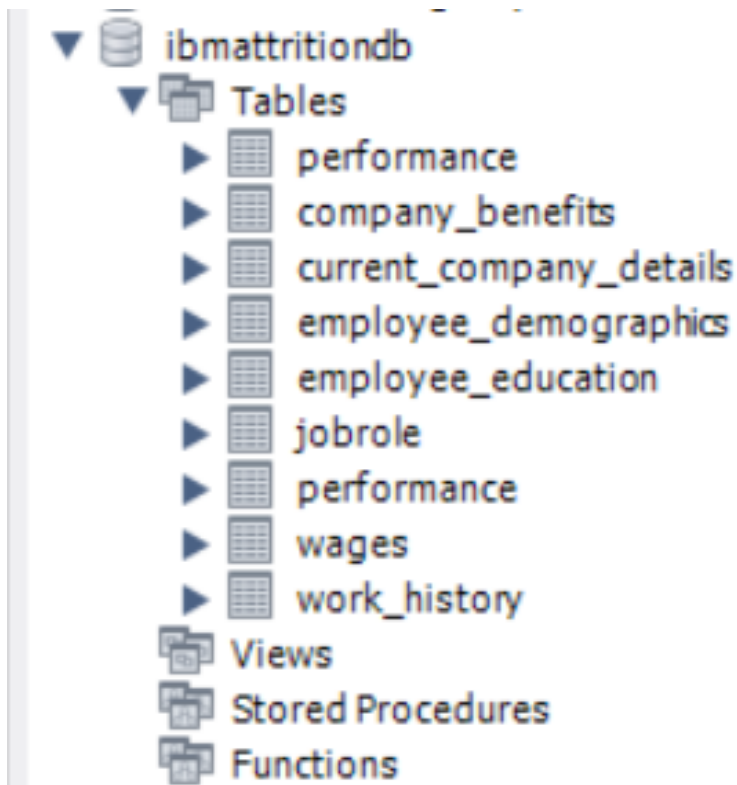
There are no empty fields or sparse data in the given dataset.

The columns/fields that we excluded from the dataset model are:

- EmployeeCount: Value is redundant with 1
- Over18: This column is excluded as it can be obtained by performing comparative operations on the Age column i.e., if Age>18, then YES.
- StandardHours: Value is redundant with 1

The above mentioned three attributes have been dropped from the 'employee_demographics' table.

Screen shot of Physical Database objects



Data in the Database

Table Name	Primary Key	Foreign Key	# of Rows in Table
employee_demographic	EmployeeNumber		1470
employee_education	EmployeeNumber	EmployeeNumber	1470
company_benefit	EmployeeNumber	EmployeeNumber	1470
current_company_archives	EmployeeNumber	EmployeeNumber	1470
wages	EmployeeNumber	EmployeeNumber	1470
jobrole_dimensions	JobLevel JobRole	JobLevel JobRole	27
performance	PercentSalaryHik	PercentSalaryHike	20
Work_history	EmployeeNumber	EmployeeNumber	1470

SQL Queries

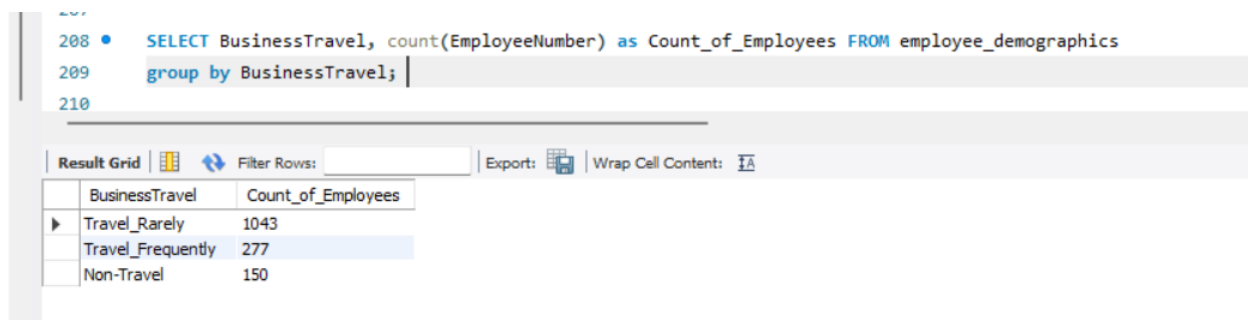
Pick 5 out of the 12 statements to write your queries in mySQL.

SQL Query 1

Assume that the company has several branch offices around the country and employee morale is down. The company feels that not enough employees are traveling frequently between offices and that more employees should travel frequently to the branch offices to improve morale amongst each other. Is the company correct in feeling this way?

We considered how many employees were travelling frequently, travelling rarely, non-travel and then we grouped the data according to Employee demographics table(Business Travel column) and counted how many employees fall into each of those 3 categories based on their travelling habits. So we can conclude that the companies feeling that not enough employees are traveling frequently between offices and that more employees should travel frequently to the branch offices to improve morale amongst each other is correct.

Screen Shot of SQL Query and Results



The screenshot shows a SQL query editor with the following code:

```
208 • SELECT BusinessTravel, count(EmployeeNumber) as Count_of_Employees FROM employee_demographics
209 group by BusinessTravel;
210
```

Below the query editor is a results grid with the following data:

BusinessTravel	Count_of_Employees
Travel_Rarely	1043
Travel_Frequently	277
Non-Travel	150

SQL Query 2

Question

An employee in Sales department has complained to HR saying that females are paid less than males in the company, in all departments. What insight can you provide to prove or disprove that statement?

We disapprove the statement because female employees have been getting lower salary than male employees only in the sales department. Whereas in other departments like Human Resources and Research & Development female employees are getting comparatively more salary than male employees.

Screen Shot of SQL Query and Results

The screenshot shows a SQL IDE interface with a query editor and a results grid. The query is as follows:

```
1 • use ibmattritiondata;
2 • select Gender,Department,avg(MonthlyIncome)
3   from jobrole j inner join employee_demographics ed
4   on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel
5   inner join employee_education e on e.employee_demographics_EmployeeNumber = ed.EmployeeNumber
6   inner join wages w on w.EmployeeNumber = ed.EmployeeNumber
7   group by Gender,Department
8   order by Department
```

The results grid displays the following data:

	Gender	Department	avg(MonthlyIncome)
▶	Male	Human Resources	4100.2222
	Female	Human Resources	4540.6875
	Male	Research & Development	6177.4254
	Female	Research & Development	6547.5303
	Female	Sales	7137.4922
	Male	Sales	7142.8175

The interface includes a toolbar with icons for file operations, a 'Limit to 5000 rows' dropdown, and a 'Result Grid' button. The results grid has a 'Filter Rows' input field, an 'Export' button, and a 'Wrap Cell Content' checkbox. The status bar at the bottom indicates 'Result 2' and 'Read Only'.

SQL Query 3

Question

A press article in a business magazine has said that at this company, married men have higher performance ratings than divorced or single men. What initial finding can you obtain from the data to help articulate the company's response in this regard?

The statement is True, married men were getting comparatively more higher ratings(4) than divorced and single men.

Screen Shot of SQL Query and Results

The screenshot shows a SQL IDE interface with a query editor and a results grid. The query is as follows:

```
1 • use ibmattritiondata;  
2 • select Maritalstatus, count(ed.EmployeeNumber) as Count_of_Employees_with_Highest_Rating  
3 • from employee_demographics ed join wages w on ed.EmployeeNumber=w.EmployeeNumber  
4 • where PerformanceRating =4 and Gender = 'Male'  
5 • group by Maritalstatus;
```

The results grid displays the following data:

Maritalstatus	Count_of_Employees_with_Highest_Rating
Married	59
Divorced	29
Single	44

The interface includes a toolbar with various icons, a "Limit to 5000 rows" dropdown, and a "Read Only" status indicator at the bottom right.

SQL Query 4

Question

An employee from Medical education field working in Sales department has spread a rumor saying that employees with his educational background are paid more in Research & Development than in Sales. What insight can you provide to prove or disprove that statement?

The statement is False, because employees with medical background have been paid higher in Sales department (average income is approx. 7.5k) compared to Research & Development department(average income is approx. 6.3k).

Screen Shot of SQL Query and Results

The screenshot shows a SQL IDE interface with a query editor and a results grid. The query is as follows:

```
1 • use ibmattributiondata;  
2 • select Department, EducationField, avg(MonthlyIncome)  
3 from jobrole j inner join employee_demographics ed  
4 on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel  
5 inner join employee_education e on e.employee_demographics_EmployeeNumber = ed.EmployeeNumber  
6 inner join wages w on w.EmployeeNumber = ed.EmployeeNumber  
7 where EducationField="Medical"  
8 group by Department, EducationField;
```

The results grid displays the following data:

Department	EducationField	avg(MonthlyIncome)
Research & Development	Medical	6318.4704
Sales	Medical	7413.4082
Human Resources	Medical	4644.1818

The interface includes a toolbar with various icons, a 'Limit to 5000 rows' dropdown, and a 'Read Only' status indicator at the bottom right.

SQL Query 5

Question

A new employee from a Life Sciences education field wants to work in Sales. Do you believe the company might be able to give him a chance to work in Sales? Why or Why not?

Only 4 % of people with Life Science background are working in sales department . Yet, there is a chance for the employee from Life Science education background at sales because all 17 of them had highest performance rate (4).

Screen Shot of SQL Query and Results

The screenshot shows a SQL query editor with the following query:

```
1 • use ibmattributiondata;
2 • select Department, EducationField, count(ed.EmployeeNumber), count(PerformanceRating),
3 • (select count(ed.EmployeeNumber) from jobrole j inner join employee_demographics ed
4 • on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel
5 • where Department='Sales') as Total_Employees_in_Sales
6 • from jobrole j inner join employee_demographics ed
7 • on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel
8 • inner join employee_education e on e.employee_demographics_EmployeeNumber = ed.EmployeeNumber
9 • inner join wages w on w.EmployeeNumber = ed.EmployeeNumber
10 • where EducationField= 'Life Sciences' and Department='Sales' and PerformanceRating=4
11 • group by Department, EducationField;
```

Below the query editor, the results are displayed in a table with the following columns: Department, EducationField, count(ed.EmployeeNumber), count(PerformanceRating), and Total_Employees_in_Sales. The table contains one row of data:

Department	EducationField	count(ed.EmployeeNumber)	count(PerformanceRating)	Total_Employees_in_Sales
Sales	Life Sciences	17	17	456

The interface also includes a toolbar with various icons, a 'Filter Rows' section, an 'Export' button, and a 'Wrap Cell Content' option. On the right side, there are buttons for 'Result Grid', 'Form Editor', and 'Read Only'.

SQL Query 6

A new employee from a Medical-related education field wants to work in Sales. Do you believe the company might be able to give her a chance to work in Sales? Why or Why not?

Screen Shot of SQL Query and Results

The screenshot shows an SQL IDE interface with a query editor and a results grid. The query editor contains the following SQL code:

```
1 • use ibmattritiondata;  
2 • select Department, EducationField, count(ed.EmployeeNumber), count(PerformanceRating),  
3 (select count(ed.EmployeeNumber) from jobrole j inner join employee_demographics ed  
4 on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel  
5 where Department='Sales') as Total_Employees_in_Sales  
6 from jobrole j inner join employee_demographics ed  
7 on j.Jobrole=ed.Jobrole and j.Joblevel=ed.Joblevel  
8 inner join employee_education e on e.employee_demographics_EmployeeNumber = ed.EmployeeNumber  
9 inner join wages w on w.EmployeeNumber = ed.EmployeeNumber  
10 where EducationField= 'Medical' and Department='Sales' and PerformanceRating=4  
11 group by Department, EducationField;
```

The results grid displays the following data:

Department	EducationField	count(ed.EmployeeNumber)	count(PerformanceRating)	Total_Employees_in_Sales
Sales	Medical	19	19	456

The interface includes a toolbar with various icons, a "Limit to 5000 rows" dropdown, and a "Read Only" status indicator at the bottom right.

Data Review for MongoDB

In MongoDB we made some changes to our table structure for feasibility.

Many of the redundant columns were dropped and all the columns required for our queries are combined in one table. Employee_data is our collection containing all the required columns and documents. Mongo database does not have to be in 3NF.

All the queries in the document are able to provide solutions or explanations to Business specific queries which are closely aligned with the real-world scenarios.

Screen shot of Physical Database objects (Database, Collections and Attributes)

The screenshot displays the MongoDB Atlas interface for the `ibm_attrition.employee_data` collection. The top right corner shows **1.5k** documents and **1** index. The **Documents** tab is selected, showing a search bar with the query `{ field: 'value' }` and buttons for **Reset**, **Find**, and **More Options**. Below the search bar are buttons for **ADD DATA** and **EXPORT COLLECTION**. The document list shows 1 - 20 of 1470 documents. The first document is expanded, revealing the following fields and values:

```
{
  "_id": ObjectId('6395dbc5acd16937790e1592'),
  "Age": 41,
  "Attrition": "Yes",
  "BusinessTravel": "Travel_Rarely",
  "DailyRate": 1102,
  "Department": "Sales",
  "DistanceFromHome": 1,
  "Education": 2,
  "EducationField": "Life Sciences",
  "EmployeeCount": 1,
  "EmployeeNumber": 1,
  "EnvironmentSatisfaction": 2,
  "Gender": "Female",
  "HourlyRate": 94,
  "JobInvolvement": 3,
  "JobLevel": 2,
  "JobRole": "Sales Executive",
  "JobSatisfaction": 4,
  "MaritalStatus": "Single",
  "MonthlyIncome": 5993,
  "MonthlyRate": 19479
}
```


MongoDB Queries/Code

Mongo Query 1

Question

If the company wants to cut travel costs, which department should the company focus on?

Research & Development Department travels more frequently than others when compared to Sales and Human Resources. So, if the company wants to cut travel costs then the company should focus on Research & Development Department.

Screen Shot of MongoDB Query/Code and Results

```
>_MONGOSH
> db.employee_data.aggregate([{$match:{"BusinessTravel":"Travel_Frequently"}},{$group:{_id:"$Department",count:{$sum:1}}}]
< { _id: 'Sales', count: 84 }
  { _id: 'Human Resources', count: 11 }
  { _id: 'Research & Development', count: 182 }
ibm_attrition>
```

Mongo Query 2

Question

HR feels that their environment satisfaction score is higher than Sales but HR job satisfaction score is lower than Research & Development. Are they right?

They are right, as we can see from the query output environment satisfaction score of HR is 2.68 which is almost equal and higher than Sales (2.67). The job satisfaction score of HR is 2.60 which is lower than Research & Development(2.72).

Screen Shot of MongoDB Query/Code and Results

```
_MONGOSH
db.employee_data.aggregate([{$match:{Department:{$in:["Sales","Research & Development","Human Resources"]}}},{ $group:{_id:"Department",avg_ES:{$avg:"$EnvironmentSatisfacti
{ { _id: 'Sales',
  avg_ES: 2.679372197309417,
  avg_JS: 2.7511210762331837 }
{ _id: 'Research & Development',
  avg_ES: 2.7440166493236213,
  avg_JS: 2.7263267429760667 }
{ _id: 'Human Resources',
  avg_ES: 2.6825396825396823,
  avg_JS: 2.6031746031746033 }
```

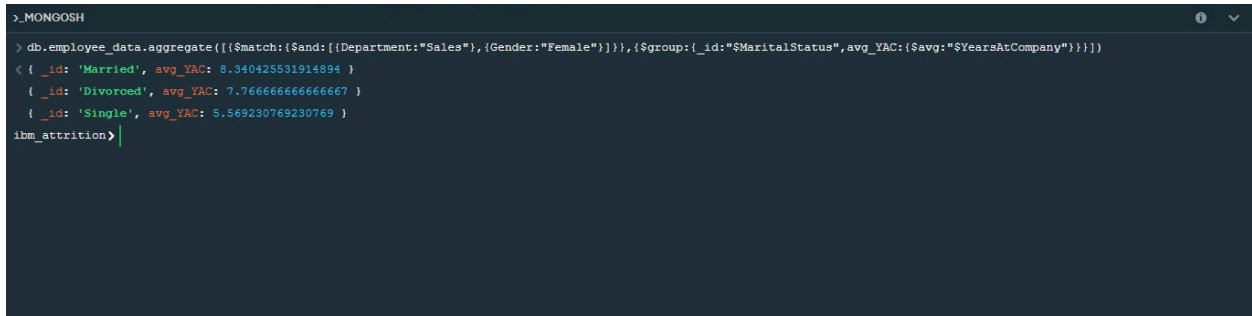
Mongo Query 3

Question

A press article in a business magazine has said that at this company, single women in Sales have worked at the company longer than divorced or married women. What initial finding can you obtain from the data to help articulate the company's response in this regard?

The press article in the business magazine is not True. The average Years at Company for single women is 5.56 which is less than Married and Divorced women 8.3 and 7.7 respectively. So the companies response is not in accordance with the business magazine.

Screen Shot of MongoDB Query/Code and Results



```
> MONGODB
> db.employee_data.aggregate([{$match:{$and:([Department:"Sales"),(Gender:"Female")]}},{ $group:{_id:"$MaritalStatus",avg_YAC:{$avg:"$YearsAtCompany"}}}])
{ "_id": "Married", avg_YAC: 8.340425531914894 }
{ "_id": "Divorced", avg_YAC: 7.766666666666667 }
{ "_id": "Single", avg_YAC: 5.569230769230769 }
ibm_attrition>
```

The screenshot shows a MongoDB shell session. The user enters a query to aggregate data from the 'employee_data' collection, filtering for Sales department females and grouping by marital status to calculate average years at company. The results show that married women have the highest average years at company (8.34), followed by divorced women (7.77), and single women have the lowest average years at company (5.57).